Nova AI Assistant

Command-Based CLI System

Course: Advanced Programming (PCCCS495)

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Semester: 4th

Introduction & Problem Statement

What is CLI

Command Line Interface enables direct text-based system interaction

Why Efficiency Matters

Text commands execute faster than GUI clicks for experienced users

Limitations of GUI

Graphical interfaces consume more resources and require mouse input

Lightweight Assistant Need

Need efficient, keyboard-driven tool for quick tasks without overhead

Project Objectives

Modular Design

Build extensible CLI assistant with independent command modules

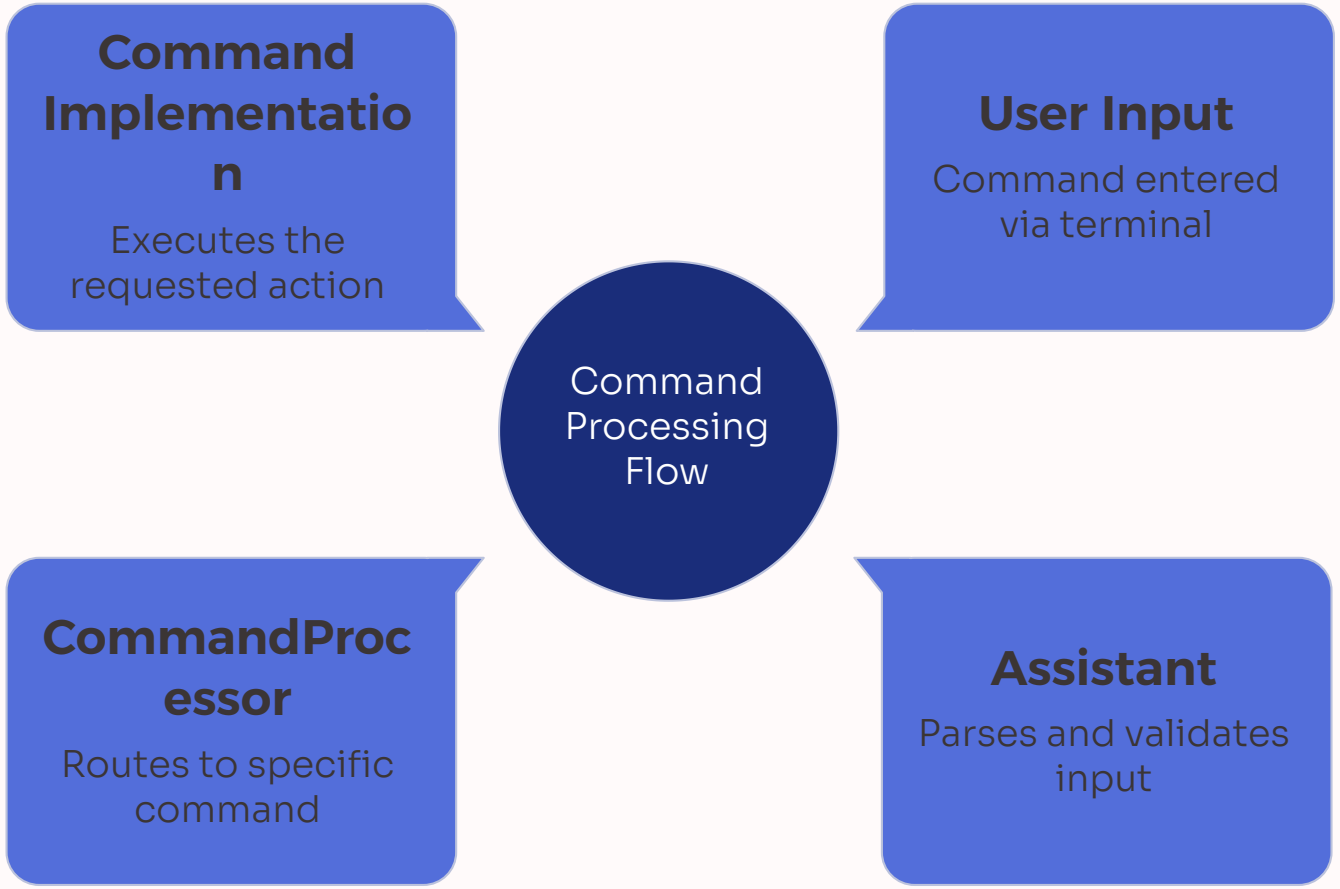
OOP Principles

Apply encapsulation, inheritance, and polymorphism throughout system

Useful Commands

Implement practical features for daily productivity needs

System Architecture



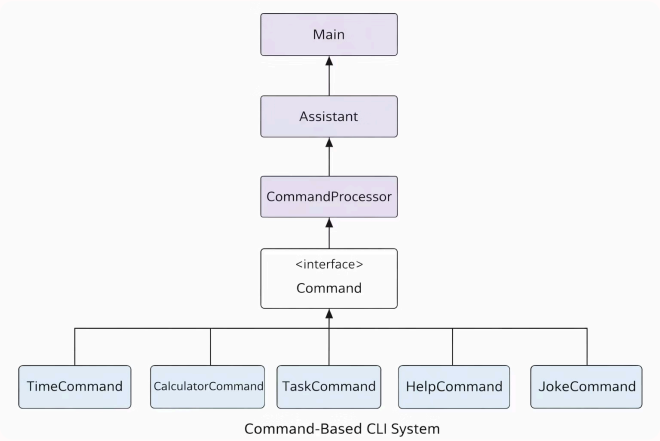
UML Class Diagram

Core Structure

- **Assistant:** Main interface
- **CommandProcessor:** Router
- **Command:** Interface

Implementation

- **TimeCommand:** System time
- **CalcCommand:** Math operations
- **TaskCommand:** Task list



Classes follow **Single Responsibility Principle** with clear separation of concerns

Key Features

Time Command

Displays current system date and time instantly

Calculator

Performs basic arithmetic operations on demand

Task Management

Add, list, and remove tasks from todo list

Help System

Lists all available commands with usage syntax

Joke Command

Returns random programming jokes for entertainment

Object-Oriented Programming Concepts

Encapsulation

Data and methods bundled within classes with private attributes

Abstraction

Complex logic hidden behind simple command interface

Inheritance

Commands extend base Command class for consistency

Polymorphism

Each command implements execute() differently

Interface Usage

Command interface ensures consistent method structure

Conclusion & Future Scope

Project Summary

Built functional CLI assistant applying core OOP principles. System demonstrates **Command Pattern** with modular, extensible architecture.

Successfully implemented time, calculator, task management, help, and joke commands.

Future Enhancements

- **GUI Interface:** Desktop application version
- **Voice Commands:** Speech recognition integration
- **AI Integration:** Machine learning for smarter responses
- **Cloud Sync:** Data synchronization across devices